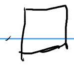
$\underline{\underline{e}} := \underline{\nabla}^s \underline{u}$

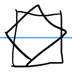
$$e_{ij} = \frac{1}{2}\left[\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}\right].$$

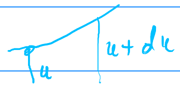
$$e_{xx} = \frac{1}{2}\left[\frac{\partial u_x}{\partial x} + \frac{\partial u_x}{\partial x}\right] = \frac{\partial u_x}{\partial x}$$

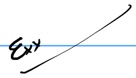
$$e_{yy} = \frac{\partial u_x}{\partial y}$$

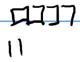
$$e_{xy} = \frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right)$$



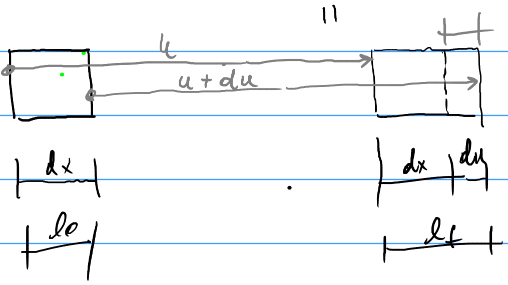




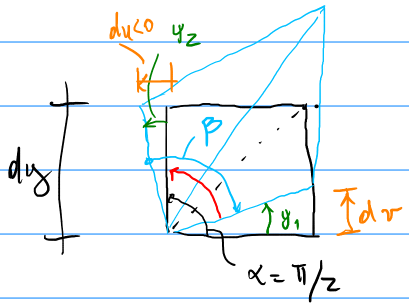




$$\frac{l_f - l_0}{l_0}$$

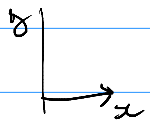


$$\varepsilon_{xx} = \frac{du}{dx}$$



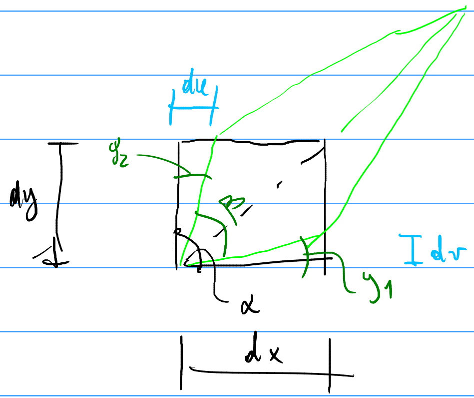
$$\begin{cases} \alpha = \pi/2 \\ \beta = \frac{\pi}{2} - \gamma_1 + \gamma_2 \end{cases}$$

$$\alpha - \beta = \gamma_1 - \gamma_2 = \frac{dv}{dx} + \frac{du}{dy} = \underbrace{2\,\varepsilon_{xy}}_{}^{\gamma_{xy}}$$



$$\gamma_1 \approx tg\,\gamma_1 = \frac{dv}{dx}$$

$$\gamma_2 \approx tg\,\gamma_2 = \frac{-du}{dy}$$

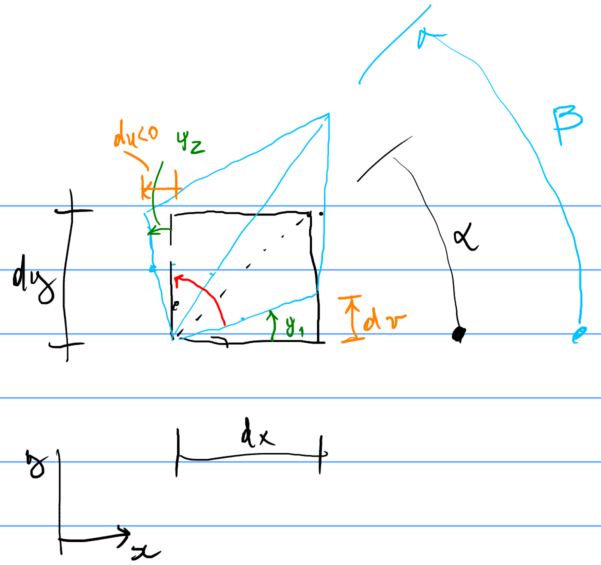


$$\left.\begin{array}{l} \alpha = \pi/2 \\ \beta = \pi/2 - \gamma_1 - \gamma_2 \end{array}\right\} \alpha - \beta = \gamma_1 + \gamma_2 =$$

$$= \frac{dv}{dx} + \frac{du}{dy} = 2\varepsilon_{xy}$$

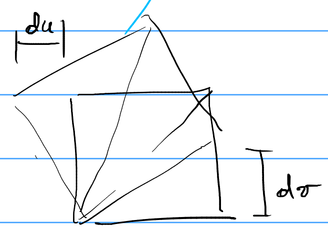
$$\gamma_2 \simeq tg(\gamma_2) = \frac{du}{dy}$$

$$\gamma_1 \simeq tg(\gamma_1) = \frac{dv}{dx}$$



$$\beta - \alpha = \frac{1}{2}\left(\gamma_1 + \cancel{\frac{\pi}{2}} + \gamma_2\right) - \cancel{\frac{1}{2}\frac{\pi}{2}} = \frac{1}{2}(\gamma_1 + \gamma_2)$$

$$\begin{cases} \alpha = \frac{1}{2}\frac{\pi}{2} \\ \beta = \frac{1}{2}\left(\gamma_1 + \frac{\pi}{2} + \gamma_2\right) \end{cases}$$



$$\begin{bmatrix} 0 & -\omega_{12} \\ \omega_{12} & 0 \end{bmatrix} \qquad \omega_{12} = \frac{1}{2}(\gamma_1 + \gamma_2) = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

Si $du = dv \Rightarrow$ la diagonal no se mueve $\Rightarrow \omega_{12} = 0$

$$A = -A^T \qquad \omega_{11} = -\omega_{11} \Rightarrow \omega_{11} = 0 \qquad \omega_{12} = -\omega_{21}$$

$$\omega_{22} = 0 \qquad \omega_{21} = -\omega_{12}$$

$$\begin{bmatrix} \omega_{11} & \omega_{12} \\ \omega_{21} & \omega_{22} \end{bmatrix} = \begin{bmatrix} -\omega_{11} & -\omega_{21} \\ -\omega_{12} & -\omega_{22} \end{bmatrix}$$

$$(\nabla \times u)_i = \varepsilon_{ijk} \frac{\partial u_k}{\partial x_j}$$

$$(\nabla \times u)_3 = \cancel{\varepsilon_{311}}^{0} \frac{\partial u_1}{\partial x_1} + \overbrace{\varepsilon_{321}}^{-1} \frac{\partial u_1}{\partial x_2} + \cancel{\varepsilon_{331}}^{0} \frac{\partial u_1}{\partial x_3} +$$

$$+ \overbrace{\varepsilon_{312}}^{1} \frac{\partial u_2}{\partial x_1} + \cancel{\varepsilon_{322}}^{0} \frac{\partial u_2}{\partial x_2} + \cancel{\varepsilon_{332}}^{0} \frac{\partial u_2}{\partial x_3} +$$

$$+ \cancel{\varepsilon_{313}}^{0} \frac{\partial u_3}{\partial x_1} + \cancel{\varepsilon_{323}}^{0} \frac{\partial u_3}{\partial x_2} + \cancel{\varepsilon_{333}}^{0} \frac{\partial u_3}{\partial x_3} = \frac{\partial u_2}{\partial x_1} - \frac{\partial u_1}{\partial x_2}$$

$$\omega_{ij} = \frac{1}{2}\left(\frac{\partial u_j}{\partial x_i} - \frac{\partial u_i}{\partial x_j}\right)$$

$$\omega_k = \tfrac{1}{2}\epsilon_{kij}\omega_{ij},$$

$$\omega_{ij} = \epsilon_{ijk}\omega_k.$$

$$\omega_i = (\nabla \times u)_i = \varepsilon_{ijk} \cdot \frac{\partial u_k}{\partial x_j}$$

$$\omega_{mn} = \varepsilon_{mni}\,\omega_i = \overbrace{\varepsilon_{mni}}^{\varepsilon_{imn}}\, \varepsilon_{ijk} \frac{\partial u_k}{\partial x_j}$$

$$= (\delta_{mj}\delta_{nk} - \delta_{mk}\delta_{nj}) \cdot \frac{\partial u_k}{\partial x_j} =$$

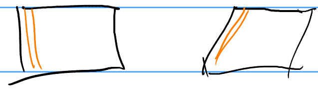
$$\omega_{mn} = \frac{\partial u_n}{\partial x_m} - \frac{\partial u_m}{\partial x_n}$$

$x_1 = a_1 + 0.01a_2$, $a_1 = x_1 - 0.01x_2$, $x_2 = a_2$, $x_3 = a_3$.

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \rightarrow \begin{cases} \varepsilon_{xx} = \dfrac{\partial u_x}{\partial x} \\ \varepsilon_{yy} = \dfrac{\partial u_y}{\partial y} \\ \varepsilon_{xy} = \dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial y} + \dfrac{\partial u_y}{\partial x}\right) \end{cases}$$

$$u_i = x_i - a_1 \Rightarrow u_1 = 0{,}01\, a_2$$

$$u_2 = 0$$



$$\begin{bmatrix} 0 & 0{,}005 \\ 0{,}005 & 0 \end{bmatrix}$$

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x} = \frac{\partial u_1}{\partial a_1} = 0$$

$$\varepsilon_{yy} = \frac{\partial u_y}{\partial y} = \frac{\partial u_2}{\partial a_2} = 0$$

$$\varepsilon_{xy} = \frac{1}{2}\left(\frac{\partial u_1}{\partial a_2} + \frac{\partial u_2}{\partial a_1}\right) = \frac{1}{2}\left(0{,}01 + 0\right) = 0{,}005$$